

Human–AI Productivity Paradoxes

A step-by-step treatment of deterministic choice and endogenous skill

Tutorial Sections 3 and 4

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Teaching slides for Aouad, Lykouris, and Zhong (2026)

Local gains can create long-run losses

At a fixed skill level

More AI never lowers optimized state-level productivity:

$$\frac{\partial p^*(s, a)}{\partial a} \geq 0.$$

After skill responds

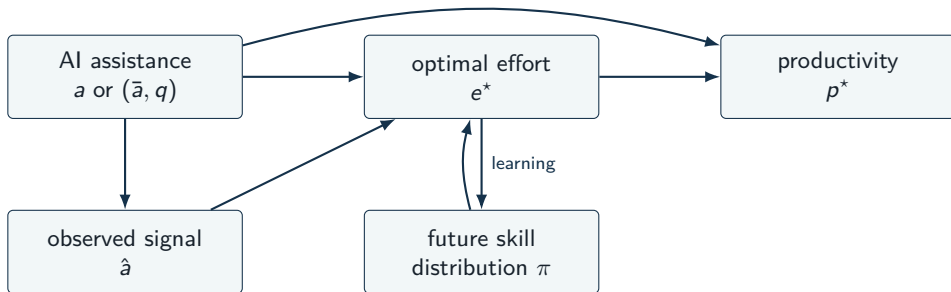
AI crowds out effort. If effort drives learning, the long-run skill distribution may shift downward:

$$a \uparrow \Rightarrow e^* \downarrow \Rightarrow \lambda(e^*) \downarrow \Rightarrow \pi \text{ shifts down.}$$

When can the second force dominate the first?

Source: Aouad, Lykouris, and Zhong (2026), Sections 2–3.

The paper separates three feedback mechanisms



- **Deterministic substitution:** AI replaces effort one for one until effort reaches zero.
- **Endogenous skill:** lower effort reduces upward skill-transition rates.
- **Information and reliability:** later sections add random AI and skill-dependent verification.

Scope

We prove the first two mechanisms completely: tutorial Sections 3 and 4.

Main conclusions before the mathematics

1. Exact static optimizer:

$$e^*(s, a) = (x^* - s - a)_+, \quad p^*(s, a) = \max\{p(x^*), p(s + a)\}.$$

2. **Skill degradation:** higher AI lowers effort-dependent upward rates, so its stationary skill distribution is first-order stochastically lower.
3. **Non-monotone productivity:** within an activity regime, long-run productivity is increasing or increasing-then-decreasing.
4. **Sensitivity gap:** for sufficiently rapid skill decay, $\Delta_m > 0$ creates a declining region.
5. **Composition reversal:** every fixed-skill productivity can rise while the stationary average falls.

Source: Paper Propositions 1 and 3, Theorem 1, and mathematical appendices.

What students should be able to reconstruct

Static choice

- ▶ Change variables from effort to total input.
- ▶ Derive the positive-part optimizer.
- ▶ Separate utility from productivity.
- ▶ Explain the kink at zero effort.

Dynamic skill

- ▶ Interpret rates and detailed balance.
- ▶ Normalize the stationary distribution.
- ▶ Prove stochastic skill degradation.
- ▶ Decompose $\mathcal{P}'(a)$ and prove one crossing.
- ▶ Recover Δ_m and the two-state threshold.

Reading rule

Every result will be obtained from the preceding equations; products, reciprocals, derivatives, and normalizations are unpacked when first used.

Part I

The deterministic basic model

Skill and AI set a baseline; effort fills the remaining productive gap.

Three inputs, two economic outcomes

Total productive input

$$x = s + e + a$$

- ▶ $s \geq 0$: current human skill
- ▶ $e \geq 0$: costly human effort
- ▶ $a \geq 0$: deterministic AI assistance

Utility and productivity

$$u(e; s, a) = p(s + e + a) - \gamma e$$
$$p^*(s, a) = p(s + e^* + a)$$

Do not subtract effort cost from productivity

The cost γe enters choice and utility, but the paper evaluates gross productive output $p(x)$.

Assumptions make the problem globally tractable

- ▶ p is nonnegative, increasing, continuous, differentiable where required, and concave.
- ▶ Effort has linear cost $c(e) = \gamma e$ with $\gamma > 0$.
- ▶ The agent is myopic: current effort does not internalize future skill.
- ▶ The maximization problem has a finite largest solution.

Define

$$g(x) := p(x) - \gamma x.$$

Because p is concave and $-\gamma x$ is linear, g is concave.

Largest maximizer

$$x^* := \max \arg \max_{x \geq 0} \{p(x) - \gamma x\}.$$

The convention makes all later formulas single valued.

Step 1: replace effort by total input

The original problem is

$$\max_{e \geq 0} \{p(s + e + a) - \gamma e\}.$$

Set

$$x = s + e + a \quad \Longleftrightarrow \quad e = x - s - a.$$

The constraint $e \geq 0$ becomes $x \geq s + a$. Substitution gives

$$\begin{aligned} p(s + e + a) - \gamma e &= p(x) - \gamma(x - s - a) \\ &= \underbrace{p(x) - \gamma x}_{g(x)} + \underbrace{\gamma(s + a)}_{\text{constant in } x}. \end{aligned}$$

Equivalent problem

$$\max_{x \geq s+a} g(x).$$

Step 2: identify the preferred total-input target

Ignoring the lower bound $x \geq s + a$, the agent wants the largest maximizer x^* of

$$g(x) = p(x) - \gamma x.$$

For differentiable p , an interior target satisfies

$$g'(x^*) = p'(x^*) - \gamma = 0 \implies p'(x^*) = \gamma.$$

Concavity implies

$$p'(x) > \gamma \quad (x < x^*), \quad p'(x) < \gamma \quad (x > x^*),$$

with weak inequalities when p has flat portions.

Economic reading

x^* equates marginal production to the marginal cost of generating input through effort.

Step 3: impose the feasible lower bound

The feasible set is the half-line $[s + a, \infty)$.

Target remains feasible

If $s + a \leq x^*$, then x^* is feasible:

$$x^{\text{opt}} = x^*.$$

Baseline exceeds target

If $s + a > x^*$, the agent cannot reduce total input. Concavity makes g decrease to the right:

$$x^{\text{opt}} = s + a.$$

$$x^{\text{opt}} = \max\{x^*, s + a\}.$$

Case 1: effort fills the gap exactly

Suppose $s + a \leq x^*$. The target is feasible, so

$$e^* = x^* - s - a \geq 0.$$

Then

$$s + e^* + a = s + (x^* - s - a) + a = x^*.$$

One-for-one substitution

While effort is positive,

$$\frac{\partial e^*}{\partial a} = -1.$$

One more unit of AI reduces effort by one unit; total input and productivity remain unchanged.

Case 2: effort cannot be negative

Suppose $s + a > x^*$. Reaching the target would require

$$e = x^* - s - a < 0,$$

which violates $e \geq 0$. Any positive effort would push total input farther to the right, where concavity makes g weakly decreasing. Therefore

$$e^* = 0, \quad x^{\text{opt}} = s + a.$$

The corner is economically meaningful

Once skill plus AI exceeds the target, effort is exhausted and further AI raises total input directly.

Step 4: combine both cases with the positive part

For any real number z , define

$$z_+ := \max\{z, 0\} = \begin{cases} z, & z \geq 0, \\ 0, & z < 0. \end{cases}$$

The two effort cases become

$$e^*(s, a) = (x^* - s - a)_+.$$

Equivalently,

$$e^*(s, a) = \begin{cases} x^* - s - a, & s + a \leq x^*, \\ 0, & s + a > x^*. \end{cases}$$

Notation checkpoint

The subscript $+$ applies to the entire expression. It is not an additional positive term.

Step 5: evaluate productivity at optimal input

Productivity excludes effort cost:

$$p^*(s, a) = p(s + e^*(s, a) + a).$$

From the optimal total input,

$$s + e^* + a = \max\{x^*, s + a\}.$$

Therefore

$$\begin{aligned} p^*(s, a) &= p(\max\{x^*, s + a\}) \\ &= \max\{p(x^*), p(s + a)\}, \end{aligned}$$

because p is increasing.

$$p^*(s, a) = \max\{p(x^*), p(s + a)\}.$$

The deterministic optimizer has a kink

$$\frac{\partial e^*}{\partial a} = \begin{cases} -1, & a < x^* - s, \\ 0, & a > x^* - s, \end{cases}$$

$$\frac{\partial p^*}{\partial a} = \begin{cases} 0, & a < x^* - s, \\ p'(s + a) \geq 0, & a > x^* - s. \end{cases}$$

Two phases

Replacement phase: AI crowds out effort with no output gain.

Expansion phase: effort is zero and AI raises output directly.

At $a = x^* - s$, the level is continuous but the derivative formula changes.

Worked example: logarithmic production

Let

$$p(x) = \alpha \log(1 + x), \quad 0 < \gamma < \alpha.$$

The first-order condition is

$$\frac{\alpha}{1+x} = \gamma \implies x^* = \frac{\alpha}{\gamma} - 1.$$

Hence

$$e^*(s, a) = \left(\frac{\alpha}{\gamma} - 1 - s - a \right)_+,$$
$$p^*(s, a) = \alpha \log \left(1 + \max \left\{ \frac{\alpha}{\gamma} - 1, s + a \right\} \right).$$

Threshold

Effort reaches zero at $a = x^* - s = \alpha/\gamma - 1 - s$.

Numerical reading of the positive part

Suppose $x^* = 5$.

$$s = 2, a = 1$$

$$s + a = 3 < 5, \quad e^* = (5 - 2 - 1)_+ = 2.$$

Total input:

$$2 + 2 + 1 = 5 = x^*.$$

$$s = 4, a = 2$$

$$s + a = 6 > 5.$$

The raw expression is -1 , so

$$e^* = (-1)_+ = 0.$$

Total input is 6.

Section 3 checkpoint

1. Changing variables reveals that the real target is total input.
2. x^* solves the unconstrained concave problem.
3. Skill plus AI is a baseline that cannot be undone.
4. Effort fills the missing gap when the baseline is below x^* .
5. Once the baseline reaches x^* , effort is zero and further AI raises productivity.

The dynamic question

If effort is also the input into learning, what happens to future skill when AI replaces current effort?

Part II

Endogenous skill and the first paradox

Current effort determines how quickly the agent moves toward higher skill.

The feedback loop changes the static conclusion

At skill state s_k , optimal effort is

$$e_k(a) := e^*(s_k, a) = (x^* - s_k - a)_+.$$

Effort controls the upward transition rate:

$$s_k \longrightarrow s_{k+1} \quad \text{at rate} \quad \lambda(e_k(a)).$$

Skill depreciates at constant rate:

$$s_k \longrightarrow s_{k-1} \quad \text{at rate} \quad \mu > 0.$$

$$\begin{aligned} a \uparrow &\Rightarrow e_k(a) \downarrow \Rightarrow \lambda(e_k(a)) \downarrow \\ &\Rightarrow \text{less stationary mass on high skill} \\ &\Rightarrow \text{possible decline in long-run productivity.} \end{aligned}$$

What is a transition rate?

The model runs in continuous time. A rate describes how quickly events arrive; it is *not* itself a probability. If the process is at s_i and the rate of moving to s_j is r_{ij} , then for a short interval of length h ,

$$\mathbb{P}(S(t+h) = s_j \mid S(t) = s_i) = r_{ij}h + o(h), \quad i \neq j.$$

The notation $o(h)$ means

$$\frac{o(h)}{h} \longrightarrow 0 \quad \text{as } h \downarrow 0.$$

Units matter

A rate may exceed one because it is measured per unit of time. The short-interval probability is approximately rate $\times$ interval length.

Short-interval probabilities at an interior state

Write

$$\lambda_k(a) := \lambda(e_k(a)).$$

Conditional on being at an interior state s_k ,

$$\mathbb{P}(\text{move up in } [t, t+h]) = \lambda_k(a)h + o(h),$$

$$\mathbb{P}(\text{move down in } [t, t+h]) = \mu h + o(h),$$

$$\mathbb{P}(\text{remain at } s_k) = 1 - [\lambda_k(a) + \mu]h + o(h).$$

The probability of two or more moves is $o(h)$.

Boundary states

At s_1 there is no downward move. At s_N there is no upward move.

Competing clocks turn rates into probabilities

At an interior state, the upward and downward moves act like exponential clocks:

$$\text{total event rate} = \lambda_k + \mu.$$

Therefore

$$\mathbb{E}[\text{waiting time to next change}] = \frac{1}{\lambda_k + \mu},$$

$$\mathbb{P}(\text{up next}) = \frac{\lambda_k}{\lambda_k + \mu}, \quad \mathbb{P}(\text{down next}) = \frac{\mu}{\lambda_k + \mu}.$$

Example

If $\lambda_k = 0.30$ per year and $\mu = 0.10$ per year, the expected wait is 2.5 years and the next change is upward with probability 0.75.

The skill process is a birth–death chain



- ▶ Only neighboring moves are allowed.
- ▶ $\lambda(e) > 0$, $\lambda'(e) \geq 0$, and $\lambda''(e) \leq 0$.
- ▶ More effort speeds learning, but its marginal effect is diminishing.

The generator records instantaneous rates

For $k = 2, \dots, N - 1$, the generator matrix Q has

$$Q_{k,k+1} = \lambda_k, \quad Q_{k,k-1} = \mu, \quad Q_{k,k} = -(\lambda_k + \mu).$$

All other entries in row k are zero, so

$$\sum_j Q_{k,j} = 0.$$

The diagonal is not a negative probability

$Q_{k,k}$ is a bookkeeping term representing total outflow. The stationary condition is

$$\pi Q = 0.$$

Stationarity is a long-run time allocation

Fix the AI level a . Define

$\pi_k(a) :=$ long-run probability of skill state s_k .

Equivalently, $\pi_k(a)$ is the long-run fraction of time spent at s_k .

$$\pi_k(a) \geq 0, \quad \sum_{k=1}^N \pi_k(a) = 1.$$

Stationarity

The distribution does not change over time. At each state,

probability inflow per unit time = probability outflow per unit time.

Detailed balance equates flow across one edge

Consider the edge connecting s_k and s_{k+1} .

- Mass at s_k is $\pi_k(a)$ and moves upward at rate $\lambda_k(a)$:

$$\text{upward flow} = \pi_k(a)\lambda_k(a).$$

- Mass at s_{k+1} is $\pi_{k+1}(a)$ and moves downward at rate μ :

$$\text{downward flow} = \pi_{k+1}(a)\mu.$$

Detailed balance sets the opposite flows equal:

$$\pi_k(a)\lambda_k(a) = \pi_{k+1}(a)\mu.$$

Both sides have units of probability mass per unit of time.

Detailed balance gives an adjacent recursion

Because $\mu > 0$, divide by μ :

$$\pi_{k+1}(a) = \pi_k(a) \frac{\lambda_k(a)}{\mu}.$$

Divide once more by $\pi_k(a) > 0$:

$$\frac{\pi_{k+1}(a)}{\pi_k(a)} = \frac{\lambda_k(a)}{\mu}.$$

- ▶ If $\lambda_k > \mu$, stationary mass rises from s_k to s_{k+1} .
- ▶ If $\lambda_k < \mu$, stationary mass falls.
- ▶ If $\lambda_k = \mu$, the neighboring probabilities are equal.

Repeated substitution produces the product formula

$$\pi_2(a) = \pi_1(a) \frac{\lambda_1(a)}{\mu},$$

$$\pi_3(a) = \pi_2(a) \frac{\lambda_2(a)}{\mu} = \pi_1(a) \frac{\lambda_1(a)}{\mu} \frac{\lambda_2(a)}{\mu},$$

$$\pi_4(a) = \pi_1(a) \frac{\lambda_1(a)}{\mu} \frac{\lambda_2(a)}{\mu} \frac{\lambda_3(a)}{\mu}.$$

Hence

$$\pi_k(a) = \pi_1(a) \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu}.$$

Each factor corresponds to one intervening edge.

How to read the product symbol

$$\prod_{i=1}^{k-1} \frac{\lambda_i(a)}{\mu} = \frac{\lambda_1(a)}{\mu} \frac{\lambda_2(a)}{\mu} \dots \frac{\lambda_{k-1}(a)}{\mu}.$$

It multiplies the upward-to-downward rate ratio on every edge between s_1 and s_k . For $k = 1$, there are no factors:

$$\prod_{i=1}^0 \frac{\lambda_i(a)}{\mu} = 1.$$

Empty-product convention

The value one makes the general formula work at $k = 1$:

$$\pi_1(a) = \pi_1(a) \times 1.$$

Normalization determines the remaining constant

$$\begin{aligned} 1 &= \sum_{k=1}^N \pi_k(a) \\ &= \pi_1(a) + \sum_{k=2}^N \pi_1(a) \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu} \\ &= \pi_1(a) \left[1 + \sum_{k=2}^N \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu} \right]. \end{aligned}$$

Therefore

$$\pi_1(a) = \left[1 + \sum_{k=2}^N \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu} \right]^{-1}.$$

Why the exponent -1 appears

Let

$$Z(a) := 1 + \sum_{k=2}^N \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu}.$$

Normalization gave

$$1 = \pi_1(a)Z(a).$$

Divide by $Z(a) > 0$:

$$\pi_1(a) = \frac{1}{Z(a)} = Z(a)^{-1}.$$

Interpretation

$Z(a)$ is the normalization constant. Its reciprocal rescales the positive weights so that the probabilities sum to one.

Stationary outcomes are weighted averages

At state s_k :

$$\text{effort} = e_k(a), \quad \text{productivity} = p^*(s_k, a).$$

The law of total expectation gives

$$\mathcal{E}(a) = \sum_{k=1}^N \pi_k(a) e_k(a),$$

$$\mathcal{P}(a) = \sum_{k=1}^N \pi_k(a) p^*(s_k, a).$$

Reading the formula

Each state-specific outcome is multiplied by the long-run fraction of time spent in that state.

Two-state stationary example

Suppose

$$\lambda_1 = 0.30, \quad \mu = 0.10.$$

Detailed balance gives

$$\pi_1(0.30) = \pi_2(0.10) \implies \pi_2 = 3\pi_1.$$

Together with $\pi_1 + \pi_2 = 1$:

$$\pi_1 = 0.25, \quad \pi_2 = 0.75.$$

If efforts are $(2, 0.5)$ and productivities are $(10, 16)$:

$$\mathcal{E} = 0.25(2) + 0.75(0.5) = 0.875, \quad \mathcal{P} = 0.25(10) + 0.75(16) = 14.5.$$

Edge balance solves full stationarity

At an interior state s_k ,

$$\pi_{k-1}\lambda_{k-1} + \pi_{k+1}\mu - \pi_k(\lambda_k + \mu) = 0.$$

Detailed balance on the lower and upper edges gives

$$\pi_{k-1}\lambda_{k-1} = \pi_k\mu, \quad \pi_{k+1}\mu = \pi_k\lambda_k.$$

Substitute:

$$\begin{aligned} \pi_{k-1}\lambda_{k-1} + \pi_{k+1}\mu - \pi_k(\lambda_k + \mu) &= \pi_k\mu + \pi_k\lambda_k - \pi_k(\lambda_k + \mu) \\ &= 0. \end{aligned}$$

The boundary equations are the first and last detailed-balance equations.

Activity is determined by a state threshold

Recall

$$e_k(a) = (x^* - s_k - a)_+.$$

Then

$$\begin{aligned} e_k(a) > 0 &\iff x^* - s_k - a > 0 \\ &\iff a < x^* - s_k \\ &\iff s_k + a < x^*. \end{aligned}$$

Because $s_1 \leq \dots \leq s_N$,

$$x^* - s_1 \geq \dots \geq x^* - s_N.$$

Ordering implication

High-skill states stop exerting effort first. Active states always form a lower block $\{1, \dots, m\}$.

Constructing the activity interval I_m

If exactly the first m states are active in the interior, then

$$a < x^* - s_m, \quad a \geq x^* - s_{m+1}.$$

The paper defines

$$I_m = ((x^* - s_{m+1})_+, (x^* - s_m)_+], \quad m = 1, \dots, N-1,$$

with

$$I_0 = [(x^* - s_1)_+, \infty), \quad I_N = [0, (x^* - s_N)_+].$$

Meaning

I_m is the regime with m active states away from shared endpoints.

Within one regime, all formulas are smooth

For a in the interior of I_m :

$$\begin{aligned} k \leq m : \quad & e_k = x^* - s_k - a > 0, \quad p_k^* = p(x^*), \\ k \geq m + 1 : \quad & e_k = 0, \quad p_k^* = p(s_k + a). \end{aligned}$$

Therefore

$$e'_k(a) = \begin{cases} -1, & k \leq m, \\ 0, & k \geq m + 1, \end{cases} \quad \frac{d}{da} p_k^*(a) = \begin{cases} 0, & k \leq m, \\ p'(s_k + a), & k \geq m + 1. \end{cases}$$

Only at a regime boundary does one state change formula.

Example: locate all activity regimes

Let

$$x^* = 10, \quad (s_1, s_2, s_3) = (2, 6, 9).$$

The thresholds are

$$x^* - s_1 = 8, \quad x^* - s_2 = 4, \quad x^* - s_3 = 1.$$

Hence

$$I_3 = [0, 1], \quad I_2 = (1, 4], \quad I_1 = (4, 8], \quad I_0 = [8, \infty).$$

At $a = 3 \in I_2$:

$$e_1 = 5, \quad e_2 = 1, \quad e_3 = 0.$$

The sensitivity gap compares two responses

Fix an intermediate regime and define

$$a_m := x^* - s_m, \quad d_m := s_{m+1} - s_m > 0.$$

At $a = a_m$, for every active $j \leq m$,

$$e_j(a_m) = x^* - s_j - (x^* - s_m) = s_m - s_j.$$

The first inactive state has total input

$$s_{m+1} + a_m = x^* + s_{m+1} - s_m = x^* + d_m.$$

Why these substitutions matter

They explain every argument appearing in Δ_m .

A semi-elasticity is a logarithmic derivative

For any positive differentiable function f ,

$$\frac{f'(x)}{f(x)} = \frac{d}{dx} \log f(x).$$

It measures the proportional change in f caused by a one-unit change in x . For the reciprocal active-rate product

$$A_{1,m}(a) = \left[\prod_{j=1}^m \lambda(x^* - s_j - a) \right]^{-1},$$

take logarithms:

$$\log A_{1,m}(a) = - \sum_{j=1}^m \log \lambda(x^* - s_j - a).$$

Differentiate the learning sensitivity

Differentiate

$$\log A_{1,m}(a) = - \sum_{j=1}^m \log \lambda(x^* - s_j - a).$$

The derivative of $x^* - s_j - a$ is -1 , canceling the outer minus sign:

$$\begin{aligned} \frac{A'_{1,m}(a)}{A_{1,m}(a)} &= \frac{d}{da} \log A_{1,m}(a) \\ &= \sum_{j=1}^m \frac{\lambda'(x^* - s_j - a)}{\lambda(x^* - s_j - a)}. \end{aligned}$$

At $a = a_m$,

$$\frac{A'_{1,m}(a_m)}{A_{1,m}(a_m)} = \sum_{j=1}^m \frac{\lambda'(s_m - s_j)}{\lambda(s_m - s_j)}.$$

The direct channel is a productivity premium

Define the premium of the first inactive state:

$$R_m(a) := p(s_{m+1} + a) - p(x^*) > 0.$$

Because $p(x^*)$ is constant,

$$R'_m(a) = p'(s_{m+1} + a).$$

Therefore

$$\frac{R'_m(a)}{R_m(a)} = \frac{p'(s_{m+1} + a)}{p(s_{m+1} + a) - p(x^*)}.$$

At $a = a_m$,

$$\frac{R'_m(a_m)}{R_m(a_m)} = \frac{p'(x^* + d_m)}{p(x^* + d_m) - p(x^*)}.$$

Learning sensitivity minus productivity sensitivity

$$\Delta_m = \underbrace{\sum_{j=1}^m \frac{\lambda'(s_m - s_j)}{\lambda(s_m - s_j)}}_{\text{learning-weight sensitivity}} - \underbrace{\frac{p'(x^* + d_m)}{p(x^* + d_m) - p(x^*)}}_{\text{productivity-premium sensitivity}}.$$

- ▶ $\Delta_m > 0$: the deskilling channel is locally more sensitive.
- ▶ $\Delta_m < 0$: the direct productivity channel is locally more sensitive.

Clarification

The sign of Δ_m alone does not determine $\mathcal{P}'(a)$ for every μ . Decay controls how strongly these sensitivities enter stationary weights.

Numerical reading of Δ_1

Take

$$\lambda(e) = 1 + e, \quad p(x) = \log(1 + x), \quad x^* = 3, \quad s_1 = 0, \quad s_2 = 2.$$

Then $d_1 = 2$ and

$$\frac{\lambda'(0)}{\lambda(0)} = 1.$$

The productivity-premium sensitivity is

$$\frac{p'(5)}{p(5) - p(3)} = \frac{1/6}{\log(6) - \log(4)} \approx 0.411.$$

Hence

$$\Delta_1 \approx 1 - 0.411 = 0.589 > 0.$$

The learning channel is locally more sensitive.

First-order stochastic dominance: the target

Let $a_\ell < a_h$. Lower AI should produce a higher skill distribution. Define the lower-tail probability

$$F^q(k) := \sum_{j=1}^k \pi_j(a_q), \quad q \in \{\ell, h\}.$$

The low-AI distribution first-order stochastically dominates the high-AI distribution if

$$F^\ell(k) \leq F^h(k) \quad \text{for every } k.$$

Why the inequality points this way

A smaller probability of being among the lowest k states means more probability on higher skill states.

Step 1: compare every adjacent ratio

The deterministic optimizer implies

$$e_i(a_\ell) \geq e_i(a_h) \quad \text{for every } i.$$

Since λ is increasing and μ is unchanged,

$$r_i^\ell := \frac{\lambda(e_i(a_\ell))}{\mu} \geq r_i^h := \frac{\lambda(e_i(a_h))}{\mu}.$$

Detailed balance gave

$$r_i^q = \frac{\pi_{i+1}(a_q)}{\pi_i(a_q)}.$$

Meaning

Every adjacent high-to-low probability ratio is more favorable to high skill under lower AI.

Step 2: remove normalization

Define unnormalized weights

$$w_1^q = 1, \quad w_k^q = \prod_{i=1}^{k-1} r_i^q \quad (k \geq 2),$$

and

$$T^q = \sum_{j=1}^N w_j^q.$$

Then

$$\pi_k(a_q) = \frac{w_k^q}{T^q}, \quad F^q(k) = \frac{\sum_{j=1}^k w_j^q}{\sum_{j=1}^N w_j^q}.$$

The problem is now a comparison between normalized products of ordered adjacent ratios.

Step 3: cross-multiply positive denominators

Because all weights are positive,

$$F^\ell(k) \leq F^h(k) \iff \left(\sum_{j=1}^k w_j^\ell \right) \left(\sum_{n=1}^N w_n^h \right) \leq \left(\sum_{j=1}^k w_j^h \right) \left(\sum_{n=1}^N w_n^\ell \right).$$

After expansion, terms with both $j \leq k$ and $n \leq k$ cancel after interchanging the dummy indices. It remains to prove, for every cross-block pair $j \leq k < n$,

$$w_j^\ell w_n^h \leq w_j^h w_n^\ell.$$

Step 4: prove each cross-block comparison

Since $n > j$, all factors before edge j cancel:

$$\begin{aligned}\frac{w_n^\ell}{w_j^\ell} &= \frac{\prod_{i=1}^{n-1} r_i^\ell}{\prod_{i=1}^{j-1} r_i^\ell} \\ &= \prod_{i=j}^{n-1} r_i^\ell \geq \prod_{i=j}^{n-1} r_i^h = \frac{w_n^h}{w_j^h}.\end{aligned}$$

Multiplying by $w_j^\ell w_j^h > 0$ gives

$$w_j^\ell w_n^h \leq w_j^h w_n^\ell.$$

Summing over all cross-block pairs proves

$$F^\ell(k) \leq F^h(k).$$

Skill degradation proposition

Proposition

For $a_\ell < a_h$,

$$\sum_{i=1}^k \pi_i(a_\ell) \leq \sum_{i=1}^k \pi_i(a_h), \quad k = 1, \dots, N.$$

More AI lowers effort, lowers every upward rate, and shifts long-run mass toward low skill.

What remains

The proposition establishes a downward skill shift, but aggregate productivity also has a positive direct AI effect. We still need to compare magnitudes.

Three-state dominance example

Suppose

$$(r_1^\ell, r_2^\ell) = (2, 1.5), \quad (r_1^h, r_2^h) = (1, 0.75).$$

The unnormalized weights are

$$w^\ell = (1, 2, 3), \quad w^h = (1, 1, 0.75).$$

After normalization,

$$\pi(a_\ell) = \left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2} \right), \quad \pi(a_h) = \left(\frac{4}{11}, \frac{4}{11}, \frac{3}{11} \right).$$

Lower tails:

$$\frac{1}{6} < \frac{4}{11}, \quad \frac{1}{2} < \frac{8}{11}.$$

Rewrite productivity using unnormalized weights

Fix an intermediate regime I_m . Define

$$q_1(a) = 1, \quad q_k(a) = \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu}.$$

Then

$$\pi_k(a) = \frac{q_k(a)}{\sum_{j=1}^N q_j(a)}.$$

Therefore

$$\mathcal{P}(a) = \frac{\sum_{k=1}^N q_k(a) p_k^*(a)}{\sum_{k=1}^N q_k(a)}.$$

The common stationary normalization constant has canceled.

Split active and inactive weights

Inside I_m ,

$$\lambda(e_i(a)) = \begin{cases} \lambda(x^\star - s_i - a), & i \leq m, \\ \lambda(0), & i \geq m + 1. \end{cases}$$

For $k \leq m$,

$$q_k(a) = \frac{\prod_{i=1}^{k-1} \lambda(x^\star - s_i - a)}{\mu^{k-1}}.$$

For $k \geq m + 1$,

$$q_k(a) = \frac{[\prod_{i=1}^m \lambda(x^\star - s_i - a)] \lambda(0)^{k-m-1}}{\mu^{k-1}}.$$

Subtract the common active-state productivity

Let $Q(a) = \sum_k q_k(a)$. Active states produce $p(x^*)$:

$$\begin{aligned}\mathcal{P}(a) &= \frac{p(x^*) \sum_{k=1}^m q_k + \sum_{k=m+1}^N q_k p(s_k + a)}{Q(a)} \\ &= \frac{p(x^*) Q(a) + \sum_{k=m+1}^N q_k [p(s_k + a) - p(x^*)]}{Q(a)} \\ &= p(x^*) + \frac{\sum_{k=m+1}^N q_k [p(s_k + a) - p(x^*)]}{Q(a)}.\end{aligned}$$

Cancel the common active-rate product

Define

$$C_m(a) := \prod_{i=1}^m \lambda(x^\star - s_i - a) > 0.$$

For $k \leq m$,

$$\begin{aligned} \frac{q_k(a)}{C_m(a)} &= \frac{\prod_{i=1}^{k-1} \lambda(x^\star - s_i - a)}{\mu^{k-1} \prod_{i=1}^m \lambda(x^\star - s_i - a)} \\ &= \frac{1}{\mu^{k-1} \prod_{i=k}^m \lambda(x^\star - s_i - a)}. \end{aligned}$$

For $k \geq m+1$,

$$\frac{q_k(a)}{C_m(a)} = \frac{\lambda(0)^{k-m-1}}{\mu^{k-1}}.$$

Define the active and inactive blocks

$$A_{k,m}(a) := \left[\prod_{i=k}^m \lambda(x^* - s_i - a) \right]^{-1}, \quad B_{k,m} := \lambda(0)^{k-m-1}.$$

Then

$$\boxed{\mathcal{P}(a) = p(x^*) + \frac{N_m(a)}{D_m(a)},}$$

where

$$N_m(a) = \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} [p(s_k + a) - p(x^*)],$$

$$D_m(a) = \sum_{k=1}^m \frac{A_{k,m}(a)}{\mu^{k-1}} + \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} > 0.$$

Differentiate every component

The $B_{k,m}$ are constant in a , so

$$N'_m(a) = \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} p'(s_k + a).$$

Only active terms in D_m vary:

$$D'_m(a) = \sum_{k=1}^m \frac{A'_{k,m}(a)}{\mu^{k-1}}.$$

Logarithmic differentiation gives

$$A'_{k,m}(a) = A_{k,m}(a) L_{k,m}(a),$$

$$L_{k,m}(a) = \sum_{j=k}^m \frac{\lambda'(x^* - s_j - a)}{\lambda(x^* - s_j - a)} \geq 0.$$

The quotient rule separates gain and loss

$$\mathcal{P}'(a) = \frac{1}{D_m(a)^2} \left[\left(\sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} p'(s_k + a) \right) D_m(a) - N_m(a) \left(\sum_{k=1}^m \frac{A'_{k,m}(a)}{\mu^{k-1}} \right) \right].$$

- First term: direct productivity gains in inactive states.
- Second term: loss caused by endogenous changes in stationary weights.

Since $D_m(a)^2 > 0$, only the bracket determines the sign.

Rewrite the sign as benefit minus loss

Define

$$U_m(a) := \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} p'(s_k + a),$$

$$V_m(a) := \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} [p(s_k + a) - p(x^*)],$$

$$h_m(a) := \frac{D'_m(a)}{D_m(a)}.$$

Multiplying the derivative by $D_m > 0$ gives

$$D_m(a) \mathcal{P}'(a) = U_m(a) - V_m(a) h_m(a) =: G_m(a).$$

G_m and $\mathcal{P}'$ always have the same sign.

Two monotonicities are immediate

Direct marginal benefit falls

Concavity of p makes p' weakly decreasing. Positive constant weights imply

$U_m(a)$ is weakly decreasing.

The productivity premium rises

Monotonicity of p makes each

$$p(s_k + a) - p(x^*)$$

weakly increasing, so

$V_m(a)$ is weakly increasing.

It remains to show that

$$h_m(a) = D'_m(a)/D_m(a)$$

is weakly increasing.

Rewrite h_m compactly

Put

$$\bar{A}_k(a) := \frac{A_{k,m}(a)}{\mu^{k-1}}, \quad \bar{B} := \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}},$$

$$S(a) := \sum_{k=1}^m \bar{A}_k(a).$$

Then

$$D_m = S + \bar{B}, \quad D'_m = S',$$

and

$$h_m(a) = \frac{S'(a)}{S(a) + \bar{B}}.$$

Also,

$$\bar{A}'_k(a) = \bar{A}_k(a)L_k(a), \quad L_k = L_{k,m}.$$

Concavity bounds the second derivative

For $y_j(a) = x^* - s_j - a$,

$$\begin{aligned}\frac{d}{da} \left(\frac{\lambda'(y_j)}{\lambda(y_j)} \right) &= \frac{[\lambda'(y_j)]^2 - \lambda''(y_j)\lambda(y_j)}{\lambda(y_j)^2} \\ &\geq 0,\end{aligned}$$

because $\lambda > 0$ and $\lambda'' \leq 0$. Applying the product rule to

$$\bar{A}'_k = \bar{A}_k L_k$$

gives

$$\boxed{\bar{A}''_k \geq \bar{A}_k L_k^2.}$$

Cauchy–Schwarz proves $h'_m \geq 0$

Differentiate:

$$h'_m = \frac{S''(S + \bar{B}) - (S')^2}{(S + \bar{B})^2}.$$

The bound on $\bar{A}_k''$ gives

$$S''(S + \bar{B}) \geq \left(\sum_k \bar{A}_k L_k^2 \right) \left(\sum_k \bar{A}_k \right).$$

Weighted Cauchy–Schwarz gives

$$\left(\sum_k \bar{A}_k L_k \right)^2 \leq \left(\sum_k \bar{A}_k \right) \left(\sum_k \bar{A}_k L_k^2 \right).$$

Since $S' = \sum_k \bar{A}_k L_k$,

$$\boxed{h'_m(a) \geq 0.}$$

The derivative crosses zero at most once

We have shown

$$U_m \downarrow, \quad V_m \uparrow, \quad h_m \uparrow.$$

Because $V_m, h_m \geq 0$, their product increases. Hence

$$G_m(a) = U_m(a) - V_m(a)h_m(a)$$

is weakly decreasing.

Within-regime shape lemma

On I_m , stationary productivity is either

increasing everywhere

or

increasing and then decreasing.

A decreasing subregion exists exactly when the left derivative at the right endpoint is negative.

Large decay creates a small expansion parameter

Evaluate the derivative at the right endpoint

$$a_m = x^\star - s_m.$$

Set

$$z := \frac{1}{\mu}.$$

Large memory decay μ means small $z > 0$. Let $X_m(z)$ denote the square-bracketed numerator of $\mathcal{P}'(a_m)$. Every term $1/\mu^{k-1}$ becomes

$$z^{k-1}.$$

Because all sums are finite, $X_m(z)$ is a polynomial in z .

Identify the lowest power of z

In U_m and V_m , the smallest state index is $k = m + 1$:

$$\text{lowest power} = z^{(m+1)-1} = z^m.$$

In D_m and D'_m , the smallest active index is $k = 1$:

$$\text{lowest power} = z^{1-1} = 1.$$

Therefore the lowest possible order in

$$U_m D_m - V_m D'_m$$

is z^m .

The surviving marginal pair

Only the first inactive state $k = m + 1$ paired with the lowest active index $k = 1$ contributes to this coefficient.

The leading coefficient is $-C_m\Delta_m$

Because $B_{m+1,m} = 1$, the coefficient on z^m is

$$c_m = p'(s_{m+1} + a_m)A_{1,m}(a_m) - [p(s_{m+1} + a_m) - p(x^*)]A'_{1,m}(a_m).$$

At $a_m = x^* - s_m$,

$$A_{1,m}(a_m) = \frac{1}{\prod_{i=1}^m \lambda(s_m - s_i)},$$

$$\frac{A'_{1,m}(a_m)}{A_{1,m}(a_m)} = \sum_{j=1}^m \frac{\lambda'(s_m - s_j)}{\lambda(s_m - s_j)}.$$

Thus

$$\boxed{c_m = -C_m\Delta_m}, \quad C_m > 0.$$

The leading coefficient determines the sign

We have

$$X_m(z) = (-C_m \Delta_m) z^m + \text{terms of order } z^{m+1} \text{ and higher.}$$

For sufficiently small $z > 0$, the first nonzero coefficient determines the sign. Since $z = 1/\mu$,

$$\text{sgn } \mathcal{P}'_-(a_m) = -\text{sgn } \Delta_m \quad \text{for sufficiently large } \mu.$$

- ▶ $\Delta_m > 0 \Rightarrow$ negative exit derivative.
- ▶ $\Delta_m < 0 \Rightarrow$ positive exit derivative.
- ▶ $\Delta_m = 0 \Rightarrow$ higher-order terms are required.

Theorem 1: when productivity rises and falls

Theorem

There exists $\bar{\mu} > 0$ such that for every $\mu > \bar{\mu}$ and every nonempty activity interval:

1. If $1 \leq m \leq N - 1$ and $\Delta_m > 0$, then $\mathcal{P}$ increases and then decreases on I_m .
2. If $m \in \{0, N\}$ or $\Delta_m < 0$, then $\mathcal{P}$ is weakly increasing on I_m .

One threshold works for all intermediate regimes:

$$\bar{\mu} = \max_{m=1, \dots, N-1} \bar{\mu}_m.$$

There are finitely many regimes, so the maximum is finite.

The boundary regimes are direct

l_N : all states active

Effort keeps total input at x^* :

$$p_k^* = p(x^*).$$

Thus

$$\mathcal{P}(a) = \sum_k \pi_k(a) p(x^*) = p(x^*).$$

l_0 : all states inactive

Every upward rate is $\lambda(0)$, so π is fixed:

$$\mathcal{P}'(a) = \sum_{k=1}^N \pi_k p'(s_k + a) \geq 0.$$

The sensitivity gap is needed only in intermediate regimes.

Two states: solve the stationary probabilities

Let $N = 2$ and $s_1 < s_2$:

$$\pi_1 \lambda(e_1) = \pi_2 \mu, \quad \pi_1 + \pi_2 = 1.$$

From balance,

$$\pi_2 = \pi_1 \frac{\lambda(e_1)}{\mu}.$$

Substitute into normalization:

$$\pi_1 \left(1 + \frac{\lambda(e_1)}{\mu} \right) = 1.$$

Therefore

$$\pi_1(a) = \frac{\mu}{\mu + \lambda(e_1(a))}, \quad \pi_2(a) = \frac{\lambda(e_1(a))}{\mu + \lambda(e_1(a))}.$$

Two states on the intermediate regime

On

$$I_1 = (x^* - s_2, x^* - s_1],$$

state 1 is active and state 2 is inactive:

$$e_1 = x^* - s_1 - a, \quad e_2 = 0.$$

Define

$$L(a) := \lambda(x^* - s_1 - a), \quad P_0 := p(x^*), \quad P_2(a) := p(s_2 + a).$$

Then

$$\boxed{\mathcal{P}(a) = \frac{\mu P_0 + L(a)P_2(a)}{\mu + L(a)}}.$$

Differentiate the two-state quotient

$$L'(a) = -\lambda'(x^* - s_1 - a), \quad P_2'(a) = p'(s_2 + a).$$

$$\begin{aligned} \mathcal{P}'(a) &= \frac{[L'P_2 + LP_2'](\mu + L) - [\mu P_0 + LP_2]L'}{(\mu + L)^2} \\ &= \frac{\mu L'P_2 + (\mu + L)LP_2' - \mu P_0L'}{(\mu + L)^2} \\ &= \frac{(\mu + L)Lp'(s_2 + a) - \mu\lambda'(x^* - s_1 - a)[p(s_2 + a) - p(x^*)]}{(\mu + L)^2}. \end{aligned}$$

The terms $LL'P_2$ cancel in the second line.

The two-state numerator decreases in A

The positive term is

$$(\mu + L)Lp'(s_2 + a).$$

As a rises,

$$L \downarrow, \quad p'(s_2 + a) \downarrow,$$

so this product falls. The subtracted term is

$$\mu\lambda'(x^* - s_1 - a)[p(s_2 + a) - p(x^*)].$$

As effort falls, concavity of λ makes $\lambda'(e)$ weakly rise; the productivity premium also rises. Thus the subtracted product rises.

$\implies$ the numerator decreases and crosses zero at most once.

Evaluate both endpoints

Let $d = s_2 - s_1$. At $a_L = x^* - s_2$,

$$p(s_2 + a_L) - p(x^*) = 0, \quad L(a_L) = \lambda(d),$$

so the numerator is positive:

$$(\mu + \lambda(d))\lambda(d)p'(x^*) > 0.$$

At $a_R = x^* - s_1$, define

$$R := p(x^* + d) - p(x^*) > 0, \quad p'_R := p'(x^* + d).$$

The endpoint numerator is

$$H_R = (\mu + \lambda(0))\lambda(0)p'_R - \mu\lambda'(0)R.$$

Solve the exact decay threshold

Rearrange:

$$H_R = \lambda(0)^2 p'_R - \mu[\lambda'(0)R - \lambda(0)p'_R].$$

The bracket is positive exactly when

$$\frac{\lambda'(0)}{\lambda(0)} - \frac{p'_R}{R} > 0,$$

which is $\Delta_1 > 0$. Conditional on $\Delta_1 > 0$, $H_R < 0$ exactly when

$$\mu > \bar{\mu}_2 := \frac{\lambda(0)^2 p'_R}{\lambda'(0)R - \lambda(0)p'_R}.$$

Complete two-state classification

► If

$$\Delta_1 > 0 \quad \text{and} \quad \mu > \bar{\mu}_2,$$

then $\mathcal{P}$ increases and then decreases on I_1 .

► If either condition fails, $\mathcal{P}$ is weakly increasing on I_1 .

Together with

$$I_2 : \mathcal{P} \text{ is flat}, \quad I_0 : \mathcal{P} \text{ increases},$$

the first case produces a global N -shape:

$$\text{flat} \rightarrow \text{rising} \rightarrow \text{falling} \rightarrow \text{rising}.$$

Numerical two-state decay threshold

Use

$$\lambda(e) = 1 + e, \quad p(x) = \log(1 + x), \quad x^* = 3, \quad s_1 = 0, \quad s_2 = 2.$$

Then

$$R = \log(6) - \log(4) \approx 0.4055, \quad p'_R = \frac{1}{6}.$$

Because $\lambda(0) = \lambda'(0) = 1$,

$$\bar{\mu}_2 = \frac{1/6}{0.4055 - 1/6} \approx 0.698.$$

- ▶ $\mu = 1$: above the threshold; a declining subregion exists.
- ▶ $\mu = 0.5$: below the threshold; productivity remains increasing.

A productivity shortfall can be arbitrarily large

Assume $0 = s_1 < s_2 < \dots < s_N$. Choose

$$a_h = x^*, \quad a_\ell = x^* - z, \quad 0 < z \leq \min\{x^*, s_2\}.$$

At a_h , every state exerts zero effort. At a_ℓ ,

$$e_1(a_\ell) = z, \quad e_k(a_\ell) = 0 \quad (k \geq 2).$$

Construction idea

Make zero-effort learning almost impossible but make the small effort z sufficient to move stationary mass from state 1 to state 2.

Send the zero-effort learning rate to zero

At high AI, all upward rates equal $\lambda(0)$. As $\lambda(0) \rightarrow 0$,

$$\mathcal{P}(a_h) \longrightarrow p(x^*).$$

At lower AI, the unnormalized weights are

$$1, \quad \frac{\lambda(z)}{\mu}, \quad \frac{\lambda(z)}{\mu} \frac{\lambda(0)}{\mu}, \quad \dots$$

All weights beyond state 2 vanish. Writing

$$r := \frac{\lambda(z)}{\mu},$$

the limiting probabilities are

$$\pi_1 \rightarrow \frac{1}{1+r}, \quad \pi_2 \rightarrow \frac{r}{1+r}.$$

The limiting productivity ratio

State 1 produces $p(x^*)$ and state 2 produces $p(x^* + s_2 - z)$. Therefore

$$\frac{\mathcal{P}(a_h)}{\mathcal{P}(a_\ell)} \rightarrow \frac{p(x^*)}{\frac{1}{1+r} [p(x^*) + r p(x^* + s_2 - z)]}.$$

Define

$$w := \frac{r}{1+r} = \frac{\lambda(z)}{\mu + \lambda(z)}.$$

For nearly linear production, the ratio is arbitrarily close to

$$\frac{x^*}{(1-w)x^* + w(x^* + s_2 - z)}.$$

Choose parameters to obtain any target ε

For fixed z , the limiting ratio is minimized at $x^* = z$:

$$\frac{z}{(1-w)z + ws_2}.$$

For any $0 < \varepsilon \leq 1$, set

$$z = x^* = \frac{\varepsilon s_2}{4}$$

and choose learning so that

$$w \geq 1 - \frac{\varepsilon}{4}.$$

Then

$$\frac{z}{(1-w)z + ws_2} \leq \frac{\varepsilon/4}{1 - \varepsilon/4} \leq \varepsilon.$$

Interpretation

This is an existence result, not a prediction that every calibration produces a severe decline.

The Simpson reversal is a derivative decomposition

$$\mathcal{P}(a) = \sum_{k=1}^N \pi_k(a) p_k^*(a).$$

Differentiate:

$$\mathcal{P}'(a) = \underbrace{\sum_{k=1}^N \pi_k(a) \frac{\partial p_k^*(a)}{\partial a}}_{\text{within-state effect } \geq 0} + \underbrace{\sum_{k=1}^N \pi'_k(a) p_k^*(a)}_{\text{composition effect}}.$$

Because

$$\sum_k \pi'_k(a) = 0,$$

only reallocations between states with different productivity matter in the second term.

Two states reveal the negative composition term

Since $\pi'_1 = -\pi'_2$,

$$\begin{aligned}\mathcal{P}' &= \pi_1(p_1^*)' + \pi_2(p_2^*)' + \pi'_1 p_1^* + \pi'_2 p_2^* \\ &= \pi_1(p_1^*)' + \pi_2(p_2^*)' + \pi'_2(p_2^* - p_1^*).\end{aligned}$$

Skill degradation implies

$$\pi'_2(a) \leq 0,$$

while normally $p_2^* - p_1^* > 0$. Therefore

$$\pi'_2(p_2^* - p_1^*) \leq 0.$$

Aggregate productivity falls when this negative composition effect dominates the nonnegative within-state effects.

Numerical Simpson reversal

Under low AI, suppose

$$\pi^\ell = (0.25, 0.75), \quad p^\ell = (10, 16).$$

Then

$$\mathcal{P}_\ell = 0.25(10) + 0.75(16) = 14.5.$$

Under high AI, both state-level productivities rise:

$$p^h = (11, 17),$$

but deskilling changes weights to $\pi^h = (0.75, 0.25)$. Then

$$\mathcal{P}_h = 0.75(11) + 0.25(17) = 12.5 < 14.5.$$

Every group improves; the aggregate falls because composition changes.

Exogenous transitions eliminate the paradox

If upward rates are fixed numbers λ_k , then

$$\pi_k = \pi_1 \prod_{i=1}^{k-1} \frac{\lambda_i}{\mu}$$

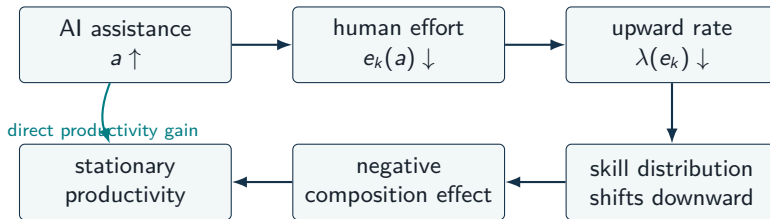
does not depend on a . Therefore $\pi'_k(a) = 0$ and

$$\mathcal{P}'(a) = \sum_{k=1}^N \pi_k \frac{\partial p_k^*(s_k, a)}{\partial a} \geq 0.$$

Identification of the mechanism

Heterogeneity alone is insufficient. The reversal requires endogenous weights: AI changes effort, effort changes learning, and learning changes long-run skill.

The complete causal chain



The theorem compares the two paths

Δ_m measures their local relative sensitivities, while μ determines how strongly the learning path influences long-run weights.

Final conclusions

1. **Static substitution is exact:**

$$e^*(s, a) = (x^* - s - a)_+.$$

2. **Fixed-skill productivity never falls:**

$$p^*(s, a) = \max\{p(x^*), p(s + a)\}.$$

3. **Endogenous learning shifts stationary mass toward low skill.**
4. **Within each activity regime, productivity has at most one interior peak.**
5. **A positive sensitivity gap and sufficiently large decay create a declining region.**
6. **The paradox is compositional, not a failure of the direct productivity effect.**

Questions that test real understanding

1. Why is $x^{\text{opt}} = \max\{x^*, s + a\}$ rather than a minimum?
2. Why may a transition rate exceed one?
3. Which normalization step produces the exponent -1 in $\pi_1(a)$?
4. Why does $r_i^\ell \geq r_i^h$ imply a lower stationary lower tail?
5. Which terms in $\mathcal{P}'(a)$ are the direct gain and composition loss?
6. Why does Cauchy–Schwarz imply at most one derivative crossing?
7. What role does μ play beyond the sign of Δ_m ?

A correct answer connects the algebra to the economic mechanism.

References and scope



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Deck scope

Tutorial Sections 3 and 4 only. Unreliable AI, AI literacy, and convex-cost extensions are outside this deck.

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